

Market Making on Binary Options

1 The problem

Write a market maker: a program that quotes both a buy and a sell price on demand, and is held to them.

Everything it trades is a **binary option** — a contract naming an event and an expiry, paying \$1 if the event happened and \$0 if not. That payoff is the whole premise, because it makes price and probability the same thing:

$$\text{value} = 1 \times \mathbb{P}(\text{event}) + 0 \times \mathbb{P}(\text{not}) = \mathbb{P}(\text{event}).$$

There is no discounting and nothing to calibrate to, so pricing *is* computing that probability, reported in $[0, 1]$ and read as dollars.

A contract is written as a set of *legs* against a *strike*. Each leg is one underlying j paired with a weight w_j , and the contract pays

$$\mathbf{1} \left\{ \sum_j w_j X_j(T) \geq K \right\},$$

where $X_j(T)$ is underlying j 's value on the expiry day T , K is the strike, and $\mathbf{1}\{\cdot\}$ equals 1 when the statement inside holds and 0 otherwise. A single-leg contract has $w = 1$ and reads “this underlying ends at or above K ”. The two-leg *spread* sets $w_{\text{AJR}} = +1$, $w_{\text{THR}} = -1$ and $K = 0$, so it reads “AJR ends above THR”. Those are the only two shapes that occur, though the code permits more.

The three underlyings split into two kinds, and the split is what the pricing method has to respect. **FED**, the fed funds rate, is a non-negative number confined to a discrete grid: it moves by whole steps of a fixed size, or not at all, and is floored at zero. **AJR** and **THR** are the valuations of two private AI labs — positive, continuous, and moving multiplicatively, so it is their *log* returns that are well behaved. Neither the rate's level nor the valuations' scale is specified in advance; both are handed over at the start of a session. Section 2 gives the exact dynamics.

Orders arrive two ways. An **RFQ** asks every maker for a two-sided quote and routes to the best price, *without saying which side it wants* — size is yours to control, direction is not. A **FOK** shows the complete order and you accept or decline — full information, no control over size.

Positions are margined at maximum loss. Each trade debits what it could cost you at worst: buying n at p debits np , selling n at p debits $n(1 - p)$. Payoffs credit only at expiry, so capital stays locked in between and closing a position does not release it. Solvency is tested end of day; going negative ends the session at zero.

Scoring: one **THEO** test supplies the true parameters and checks pricing alone; three **VERBOSE** tests pass on not crashing and not going bankrupt; sixteen **SCORED** tests give full credit for finishing top by profit, zero for bankruptcy, partial for surviving. Competing makers change between tests and are never identified.

Two tasks follow. **Price** the contracts given the parameters. And **infer** those parameters, since outside THEO they are hidden and all you get is a *warm-up history*: daily values of the three underlyings over some past window.

2 How the market moves

Pricing a contract means finding the probability of its event, and that probability depends entirely on how the underlyings evolve between now and expiry. So the process comes first; everything in the two sections after this is derived from it and from nothing else.

The exchange advances one day at a time. The rate moves first, then both companies react to it.

The rate takes one step of fixed size h up, down, or nowhere, with the odds tilted by how far it sits from a target r^* — below target it is likelier to rise, above target likelier to fall. That tilt is the entire content of the mean-reversion strength κ :

$$\begin{aligned} u(r) &= \text{clip}(p_u + \kappa(r^* - r), 0, 1), & d(r) &= \text{clip}(p_d - \kappa(r^* - r), 0, 1 - u(r)), \\ \mathbb{P}(\xi_t = +1) &= u(r_t), & \mathbb{P}(\xi_t = -1) &= d(r_t), & \mathbb{P}(\xi_t = 0) &= 1 - u(r_t) - d(r_t), \\ r_{t+1} &= \max(r_t + h \xi_t, 0), & \xi_t &\in \{+1, 0, -1\}. \end{aligned}$$

Here u and d are the probabilities of an up and a down step from level r , p_u and p_d are their baseline values before the tilt, and ξ_t is the direction actually drawn. The process is *trinomial*: staying put is the third outcome and takes whatever probability is left over, so it is never specified directly. This is also what the second clip is for — capping d at $1 - u$ rather than at 1 guarantees $u + d \leq 1$, so the leftover cannot go negative.

Note also the floor at zero in r_{t+1} : a down move from a level below h lands on 0 and therefore travels *less* than a full step. That detail matters twice later.

Each company moves by a log return — the log of tomorrow's value over today's — built from four pieces:

$$\underbrace{\log \frac{S_{t+1}^i}{S_t^i}}_{\text{today's move}} = \underbrace{\mu_i}_{\text{drift}} + \underbrace{\beta_i^r \Delta r_t}_{\text{reaction to the rate}} + \underbrace{\gamma_i Z_t}_{\text{shared shock}} + \underbrace{\varepsilon_t^i}_{\text{own shock}},$$

where S_t^i is company i 's valuation on day t ; μ_i is a constant daily drift; $\Delta r_t = r_{t+1} - r_t$ is that day's rate *change* in absolute points, and β_i^r is the company's sensitivity to it; $Z_t \sim N(0, \sigma_s^2)$ is a sector shock drawn once per day and applied to *both* companies with loadings γ_i ; and $\varepsilon_t^i \sim N(0, \sigma_i^2)$ is private to each. All draws are independent across days.

The two companies therefore share *two* common factors, and separating them is what makes the next section work. Both load on the same daily rate change Δr_t , and both are hit by the same sector shock Z_t ; only ε_t^i is private. So unconditionally their returns are correlated through both channels:

$$\text{Cov}(x_t^A, x_t^T) = \underbrace{\beta_A^r \beta_T^r \text{Var}(\Delta r_t)}_{\text{via the rate}} + \underbrace{\gamma_A \gamma_T \sigma_s^2}_{\text{via the sector shock}}.$$

Conditional on the terminal rate level, the rate channel contributes no variance. Its contribution over T days telescopes,

$$\sum_{t=0}^{T-1} \beta_i^r \Delta r_t = \beta_i^r (r_T - r_0),$$

so the intermediate path drops out and only the terminal level enters. Once that level is fixed the term is a constant, and Z_t is the only remaining source of correlation. The terminal level is not known in advance. Section 3 enumerates every level the lattice can reach and weights each by its probability; the rate contribution is constant within a single term of that sum, not overall. Section 4 estimates the sector term only after regressing the rate term out. Note also that the companies load on the rate's *change*, not its level.

Notation

Three different kinds of quantity get called a “value” in this problem, and keeping them apart matters: a *rate level* (a number like 3.25), a *company valuation* (a dollar amount), and a *price* (an option’s fair value, always in $[0, 1]$). The symbols, collected:

the rate		the companies	
r_t	rate level on day t	S_t^i	company i ’s valuation on day t
h	step size of the ladder	μ_i	daily drift
r^*	the level reversion pulls toward	β_i^r	sensitivity to the rate’s change
κ	strength of that pull	γ_i	loading on the sector shock
p_u, p_d	baseline up/down probabilities	Z_t	sector shock, $N(0, \sigma_s^2)$, shared
$u(r), d(r)$	tilted up/down probabilities at r	ε_t^i	own shock, $N(0, \sigma_i^2)$, private
ξ_t	direction drawn, $\in \{+1, 0, -1\}$	R_i	total log return over T days
Δr_t	one day’s rate change	V_i	one day’s return variance
ρ	a candidate rate level at expiry	C	one day’s covariance between the two
D	total rate move, $r_T - r_0$	$m_i(\rho)$	$\mathbb{E}[R_i]$ given the rate ends at ρ
the contract and the quote			
T	days until expiry	Φ	standard normal CDF
K	strike	$\hat{p}$	the maker’s estimated price
w_j	weight on leg j	W	half-spread

A hat always means “estimated from data” rather than known.

3 Pricing, when the parameters are known

Take the process of Section 2 as given, with every parameter known — the situation the THEO test sets up. The task is then purely computational: find the probability that the weighted sum clears the strike. The obvious approach is to simulate a few hundred thousand futures and count. The point of this section is that simulation is unnecessary, and that seeing why also tells us, in Section 4, precisely which parameters are worth estimating.

Why no simulation is needed

It *looks* path-dependent: each day’s company move depends on that day’s rate move. It is not. Sum the rate terms over T days:

$$\sum_{t=0}^{T-1} \beta_i^r \Delta r_t = \beta_i^r \sum_{t=0}^{T-1} (r_{t+1} - r_t) = \beta_i^r (r_T - r_0).$$

Everything in the middle cancels — the sum **telescopes**. Only where the rate *ends up* matters, never how it got there, so two paths landing on the same r_T leave the companies with identical distributions. The price splits in two.

Where does the rate land? Each day the rate moves $+1$, 0 , or -1 steps, so after T days its net displacement is some whole number of steps between $-T$ and $+T$. That is $2T + 1$ possible values, and the rate can only sit at $r_0 + kh$ for one of those k — fewer in practice, since the floor at zero collapses several down-paths onto the same level.

The point is what that number is *not*: there are 3^T distinct paths, but only $2T + 1$ destinations, because a path that goes up-then-down lands where one that goes down-then-up does. The rate therefore evolves on a *recombining trinomial lattice*, and its state space grows linearly in T rather than

exponentially. Pushing the probabilities forward one day at a time — each level distributing its mass to three successors — therefore gives the whole distribution exactly and cheaply, with no sampling.

Write ρ for one of those reachable levels — a candidate value of the **fed funds rate** on the expiry day, such as 3.25. It is not a price and not a company valuation: it is one possible answer to “where did the rate end up?”, so that $\mathbb{P}(r_T = \rho)$ is the probability it ended up there. Each such level is called a *node*. There are at most $2T + 1$ of them and their probabilities sum to one, so they partition every possible future.

Given that, where do the companies land? Both daily returns are linear combinations of the same three independent normal draws $Z_t, \varepsilon_t^A, \varepsilon_t^T$, so each day’s pair follows a *bivariate normal*; summing over independent days preserves that. (Marginal normality alone would not suffice — two normal variables need not be jointly normal — but a shared linear source gives it directly.) Writing R_A and R_T for the total log returns over T days and $D = r_T - r_0$ for the total rate move,

$$\begin{pmatrix} R_A \\ R_T \end{pmatrix} \Big| D \sim N \left(\begin{pmatrix} T\mu_A + \beta_A^r D \\ T\mu_T + \beta_T^r D \end{pmatrix}, T \begin{pmatrix} V_A & C \\ C & V_T \end{pmatrix} \right),$$

Here $V_i = \gamma_i^2 \sigma_s^2 + \sigma_i^2$ is one day’s return variance for company i — the shared shock’s contribution plus its own — and $C = \gamma_A \gamma_T \sigma_s^2$ is one day’s covariance between the two. Note that C carries *only* the sector term: this distribution is conditional on D , so the rate channel of Section 2 contributes to the mean rather than to the covariance, and the private shocks are independent. The mean is drift plus the rate’s pull; both variance and covariance scale by T because days are independent and variances of independent sums add.

Combining the two halves is just the law of total probability — since the nodes partition every possible future, split on which one occurs, price each case, and weight by how likely that case is:

$$\text{price} = \sum_{\rho} \mathbb{P}(r_T = \rho) \cdot \mathbb{P}(\text{contract wins} \mid r_T = \rho),$$

the sum running over every reachable node ρ .

Inside a node the rate is no longer random — it is the known constant ρ — which is what makes each term tractable. A FED leg then contributes a fixed $w_r \rho$ to the weighted sum, and a known constant can be moved to the other side of the inequality, so the rate leg just shifts the strike: $K \mapsto K - w_r \rho$. Substituting $D = \rho - r_0$ into the block above gives the companies’ distribution at that node. What remains is one of three cases.

The three inner probabilities

Rate only. Once the rate’s landing place is fixed, nothing random is left inside the node, so the contract simply wins or loses there. Summing the winning nodes’ probabilities gives the price.

One company. Write S_0^i for company i ’s value today and R_i for its total log return over the T days, so its value at expiry is $S_T^i = S_0^i e^{R_i}$ — today’s value grown by the accumulated return. The contract’s condition $w S_0^i e^{R_i} \geq K$ is then a statement about R_i alone, and taking logs turns it into a plain threshold:

$$w S_0^i e^{R_i} \geq K \iff R_i \geq \log(K / (w S_0^i)).$$

At this node R_i is normal. Its mean is the quantity

$$m_i(\rho) = \underbrace{T\mu_i}_{\text{drift, once per day for } T \text{ days}} + \underbrace{\beta_i^r(\rho - r_0)}_{\text{reaction to the total rate move } r_0 \rightarrow \rho},$$

which is simply the mean from the bivariate block above with $D = \rho - r_0$ substituted in — $m_i(\rho)$ names “company i ’s expected total log return, given the rate ends at ρ ”. It is a function of ρ because that is the only way the rate enters, which is exactly what the telescoping bought. Its variance is TV_i , which does *not* depend on ρ : the rate contributes to where the return is centred, never to how spread out it is.

So the probability wanted is $\mathbb{P}(R_i \geq c)$ with the threshold $c = \log(K/(wS_0^i))$, and $R_i \sim N(m_i(\rho), TV_i)$. The only tool needed is that subtracting a normal’s mean and dividing by its standard deviation leaves a *standard* normal, $Z \sim N(0, 1)$, whose CDF is Φ . Doing that to both sides of the inequality — which is legitimate because subtracting a constant and dividing by a positive one preserve the direction — gives

$$\mathbb{P}(R_i \geq c) = \mathbb{P}\left(\underbrace{\frac{R_i - m_i(\rho)}{\sqrt{TV_i}}}_{Z \sim N(0, 1)} \geq \frac{c - m_i(\rho)}{\sqrt{TV_i}}\right) = 1 - \Phi\left(\frac{c - m_i(\rho)}{\sqrt{TV_i}}\right) = \Phi\left(\frac{m_i(\rho) - c}{\sqrt{TV_i}}\right).$$

The second equality is the definition of the CDF — Φ gives the probability of falling *below* a point, so clearing it is one minus that — and the third uses the symmetry of the normal about zero, $1 - \Phi(z) = \Phi(-z)$, which just flips the sign of the fraction. Writing c out in full recovers the formula:

$$\mathbb{P}(R_i \geq \log(K/(wS_0^i))) = \Phi\left(\frac{m_i(\rho) - \log(K/(wS_0^i))}{\sqrt{TV_i}}\right).$$

Read plainly, the fraction is *how far the expected return sits above the threshold, measured in standard deviations*. Push the expected return up and the contract gets likelier; raise the strike and the threshold rises, so it gets less likely; increase the volatility $\sqrt{TV_i}$ and the ratio shrinks toward zero, pulling the price toward $\Phi(0) = 0.5$ — more uncertainty makes any contract closer to a coin flip.

The spread. Two-dimensional in appearance only. Asking whether one company beats the other is asking whether their *ratio* beats one, and taking logs turns that ratio into a *difference* of the two log returns:

$$w_1 S^1(T) \geq w_2 S^2(T) \iff R_1 - R_2 \geq \log \frac{w_2 S_0^2}{w_1 S_0^1}.$$

A difference of jointly normal variables is itself normal, with the variance of a difference being $\text{Var}(R_1) + \text{Var}(R_2) - 2\text{Cov}(R_1, R_2)$:

$$R_1 - R_2 \mid \rho \sim N(m_1 - m_2, T(V_1 + V_2 - 2C)),$$

so the spread reduces to the same one-dimensional Φ evaluation as a single leg. Note the $-2C$: the shared sector shock hits both companies together and cancels in the difference, which is exactly why the spread is a tighter contract than either leg alone. The weights enter only inside the threshold, so $w = 1, 2, 5, 10$ all price identically at a matched spot ratio. Only a two-leg contract at a *nonzero* strike resists this, since it is a sum of two lognormals rather than a ratio; one 1-D quadrature covers it, and it is implemented although it never occurs.

Checking it

There is no external price to check against, so what can be checked is internal consistency. The closed form and a Monte Carlo simulation are two independent implementations of the same specification,

the simulation using nothing but the template’s own step and settlement code. If the closed form is right, each $z = (\text{simulated} - \text{closed})/\text{s.e.}$ is a draw from $N(0, 1)$, and that *distribution* is the hypothesis under test — not whether any one pair of numbers looks close.

One test carries most of the weight, because it separates a real error from a lucky sample. A genuine pricing error δ produces $z = \delta/\text{s.e.}$, and the standard error shrinks as $1/\sqrt{N}$, so any real discrepancy must grow *in proportion to* the path count. Holding the scenarios fixed and varying only N :

$$\sum z^2 = 93.6 \rightarrow 108.0 \rightarrow 118.5 \quad \text{at } N = 10^4, 4 \times 10^4, 1.6 \times 10^5,$$

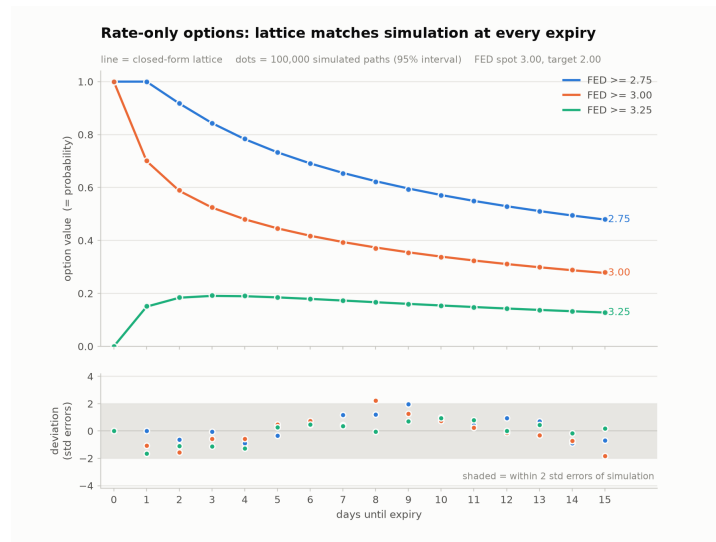
on 100 degrees of freedom. A sixteenfold increase in N would have multiplied any systematic error’s signature sixteenfold; instead all three sit within 1.4 standard deviations of the null value of 100. That is the signature of no error at all.

The rest of the check varies everything else that could conceal one. Each row reports the excess of $\sum z^2$ over its null value, in standard deviations:

varied	cases	excess (s.d.)
seed	four independent	−0.13, +0.12, +0.73, +0.96
expiry T	1–2, 3–5, 6–10, 11–20	+1.12, +2.42, +0.83, −0.76
parameter draw	narrower, lower-volatility	−1.33, −0.90, −1.45
contract family	rate, equity, scaled	+0.42, +1.42, +0.61
	negative-weight, spread	+1.39, −0.45

There is no trend in T , and turning off the scenario-selection filter changes nothing. Eighteen measurements scattered about zero with unit spread is what a correct pricer produces.

So: no detectable discrepancy, with any systematic pricing error bounded below roughly 0.05¢ — two orders of magnitude inside the estimation error of Section 4. What the check establishes remains narrow, though. The algebra matches the specification. It cannot establish that the specification was read *correctly*, because both implementations read it the same way.



Rate-only contracts, three strikes, expiries 0–15. Line is the lattice, dots are 100,000 simulated paths, lower panel the deviation in standard errors — all within ± 2 on both sides. The green curve is the tell: a strike above spot first rises as the rate gets time to reach it, then falls as mean reversion drags it back. A pricer that missed the reversion would not bend.

4 Inferring the parameters from history

Outside THEO the parameters are hidden and must come from the warm-up history. This section is not self-contained the way Section 3 was: it works only because the history is generated by *this* process, sampled daily with no gaps.

It is also not a separate problem. What has to be estimated is decided entirely by what the pricer of Section 3 actually consumes, so it is worth listing where each quantity entered there:

quantity	where it enters the price	appearing in
h , and the lines $u(\cdot), d(\cdot)$	builds the lattice, giving $\mathbb{P}(r_T = \rho)$	the outer sum
μ_i, β_i^r	centre of the return, $m_i(\rho) = T\mu_i + \beta_i^r(\rho - r_0)$	every Φ numerator
V_i	spread of the return, $\sqrt{TV_i}$	every Φ denominator
C	cancellation between the legs, $T(V_1 + V_2 - 2C)$	the spread contract only

Those are the targets, and nothing else is. The converse matters just as much: a parameter the pricer never touches on its own cannot be recovered and need not be, which is where the identifiability argument below comes from — it is a fact about the pricing formula, not about the estimator.

Why least squares, and not something cleverer

The reason is that the process of Section 2 is *already written as two regression equations*. Nothing has to be assumed or approximated; the fitting method is read off the generating process. Recall the company equation and expand the rate equation:

$$\underbrace{\log \frac{S_{t+1}^i}{S_t^i}}_{\text{outcome}} = \underbrace{\mu_i}_{\text{intercept}} + \underbrace{\beta_i^r}_{\text{slope}} \underbrace{\Delta r_t}_{\text{regressor}} + \underbrace{(\gamma_i Z_t + \varepsilon_t^i)}_{\text{mean-zero noise}}, \quad u(r) = \underbrace{(p_u + \kappa r^*)}_{\text{intercept}} \underbrace{- \kappa}_{\text{slope}} r.$$

Both have the shape $y = a + bx + \text{noise}$. So the estimator is ordinary least squares in both cases: given paired observations (x_t, y_t) it returns

$$\hat{b} = \frac{\sum_t (x_t - \bar{x})(y_t - \bar{y})}{\sum_t (x_t - \bar{x})^2}, \quad \hat{a} = \bar{y} - \hat{b} \bar{x},$$

where $\bar{x}$ and $\bar{y}$ are sample means. OLS is the right choice here rather than a convenient one: the truth is linear, the noise is additive with mean zero, and — crucially — the regressor is *exogenous*, meaning uncorrelated with the noise. That last condition holds by construction, because **advance_step** moves the rate first and only then draws Z_t and ε_t^i independently of it. With Gaussian, equal-variance noise OLS is also the maximum likelihood estimator, so there is nothing cleverer to reach for.

What changes between the two applications is only what plays x and what plays y .

Step 1: the step size h

Every rate move is $\pm h$ or zero, except at the floor where clamping shortens it. Since truncation can only ever *shorten* a move, the largest move ever observed is a full one:

$$\hat{h} = \max\{|\Delta r_t| : \Delta r_t \neq 0\}.$$

Taking the *smallest* nonzero move is the natural first guess and it is wrong — it returns the truncated stub, reporting 0.10 for a ladder whose true step is 0.25, whenever the reachable levels do not divide evenly into zero.

Step 2: the rate lines

Here $y_t = \mathbf{1}\{\Delta r_t > 0\}$, one if the rate rose that day and zero otherwise, and $x_t = r_t$, the level it started from. That pairing works because of one fact about indicators: the indicator is *Bernoulli*, and the expectation of a Bernoulli variable is the probability that it equals 1, so

$$\mathbb{E}[\mathbf{1}\{\Delta r_t > 0\} \mid r_t = r] = \mathbb{P}(\text{rate rises} \mid r) = u(r) = (p_u + \kappa r^*) - \kappa r.$$

The conditional mean of y is therefore *exactly* linear in x , which is the assumption OLS needs. The individual observations are only ever 0 or 1 and look nothing like a line, but their average height at each level traces $u(\cdot)$ precisely. (This is a linear probability model, usually a crude stand-in for a logit; here it is not an approximation at all, because the true $u(\cdot)$ really is a straight line inside the clamps. The Bernoulli noise is heteroskedastic, so OLS is unbiased but not fully efficient.)

Run the same fit on $\mathbf{1}\{\Delta r_t < 0\}$ against r_t . Comparing both fitted slopes with the expansions in Section 2,

$$b_u = -\kappa, \quad b_d = +\kappa \quad \implies \quad \hat{\kappa} = \frac{1}{2}(\hat{b}_d - \hat{b}_u),$$

averaging two independent measurements of the same quantity. One correction is needed first. At $r_t = 0$ a down draw is clamped back to 0, producing no move at all, so it is recorded as a *stay* — the down frequency is censored exactly there. Those days are excluded before fitting; otherwise, since the two slopes are averaged into one κ , the censoring would contaminate the up line as well. (The code excludes all days starting below one full step, which is conservative: at $0 < r_t < h$ a down draw still registers as a down move, merely a short one.)

Step 3: drift and rate sensitivity

The same OLS, with $y_t = \log(S_{t+1}^i/S_t^i)$ — the company's daily log return, obtained by differencing the history — and $x_t = \Delta r_t$, that day's rate change, obtained by differencing it too. Reading the coefficients off the equation above, the slope estimates β_i^r and the intercept estimates μ_i . Both companies share the same regressor Δr_t , which is why fitting them jointly as a system would return identical numbers to fitting them one at a time.

Step 4: the noise

Whatever the line failed to explain is the shock itself. Subtracting the fit leaves

$$\hat{e}_t^i = \log \frac{S_{t+1}^i}{S_t^i} - \hat{\mu}_i - \hat{\beta}_i^r \Delta r_t \approx \gamma_i Z_t + \varepsilon_t^i,$$

one draw per day of exactly the quantity whose variance and cross-moment are the targets. Since Z_t and ε_t^i are independent and mean-zero,

$$\text{Var}(\gamma_i Z_t + \varepsilon_t^i) = \gamma_i^2 \sigma_s^2 + \sigma_i^2 = V_i, \quad \text{Cov}(\gamma_A Z_t + \varepsilon_t^A, \gamma_T Z_t + \varepsilon_t^T) = \gamma_A \gamma_T \sigma_s^2 = C,$$

the covariance keeping only the shared term because the two private shocks are independent of each other and of Z_t . So the sample second moments of the residuals estimate V and C directly:

$$\hat{V}_i = \frac{1}{m-2} \sum_t (\hat{e}_t^i)^2, \quad \hat{C} = \frac{1}{m-2} \sum_t \hat{e}_t^A \hat{e}_t^T,$$

over m days. The divisor is $m-2$ rather than m because two coefficients were fitted before the residuals were formed, costing two degrees of freedom; dividing by m would understate the variance.

It matters that $\hat{C}$ comes from *residuals* rather than raw returns. The raw covariance between the two companies contains both channels of Section 2 — the shared rate change as well as the shared sector shock — and only the second belongs in the conditional distribution the pricer uses. Regressing Δr_t out first is what isolates it.

Mapping the fits back onto the parameter struct

The pricer needs a `MarketParameters` object, so the fitted coefficients have to be written back into its fields. For the companies this is immediate: $\hat{\mu}_i$ and $\hat{\beta}_i^r$ are those fields. For the other two groups it is not, and seeing why is what identifies what can be recovered at all.

The rate fields. OLS delivers two intercepts and a slope — three numbers — but the struct holds four: p_u, p_d, κ, r^* . Matching the expansions,

$$a_u = p_u + \kappa r^*, \quad a_d = p_d - \kappa r^*, \quad b_d = -b_u = \kappa.$$

Three equations, four unknowns: r^* can be moved anywhere and p_u, p_d shifted to compensate without changing $u(\cdot)$ or $d(\cdot)$ by one iota. The individual fields are therefore *not recoverable*, and it does not matter, because prices depend only on the two lines. What the code does is pick a convenient representative of that family: set $\hat{r}^* = \bar{r}$, the mean level in the history. The tilt then vanishes at $\bar{r}$, so the intercepts reduce to the fitted probabilities there — and since an OLS line passes through the sample means, those are just the observed up and down frequencies. That is exactly what the implementation stores.

The noise fields. The residual moments give three numbers (V_A, V_T, C) , but the struct holds five: $\gamma_A, \gamma_T, \sigma_s, \sigma_A, \sigma_T$. Again under-determined, and again harmlessly: scaling $\sigma_s \rightarrow c\sigma_s$ while scaling both $\gamma_i \rightarrow \gamma_i/c$ leaves V_A, V_T, C — and hence every price — unchanged. Fixing the scale at $\sigma_s = 1$ removes the redundancy, after which $\gamma_A \gamma_T = C$ and $\sigma_i^2 = V_i - \gamma_i^2$. The code splits the covariance in proportion to each name's volatility,

$$\gamma_A = \sqrt{|C|\lambda}, \quad \gamma_T = \text{sign}(C)\sqrt{|C|/\lambda}, \quad \lambda = \sqrt{V_A/V_T},$$

which reproduces $\gamma_A \gamma_T = C$ and, more to the point, keeps both idiosyncratic variances non-negative: $\gamma_A^2 = |C|\sqrt{V_A/V_T} \leq V_A$ follows from $|C| \leq \sqrt{V_A V_T}$, the Cauchy–Schwarz bound any valid covariance satisfies. Splitting the loading equally between the two names would instead require $|C| \leq \min(V_A, V_T)$ and would truncate legitimate covariances.

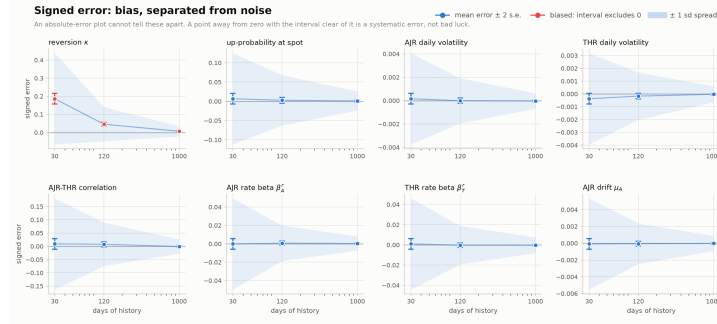
The count. Both redundancies have the same character, and both are settled by the table at the top of this section rather than by anything about the fitting. p_u and r^* reach the price only through $u(\cdot)$ and $d(\cdot)$; γ_i and σ_s reach it only through V_A, V_T, C . A quantity the pricer cannot distinguish, no estimator can distinguish either — and neither needs to.

Stripping those out leaves eleven numbers that actually determine a price: the step size h ; three for the rate lines (a_u, a_d, κ) ; μ_i and β_i^r for each of the two companies; and V_A, V_T, C . Anything beyond those is estimating an arbitrary choice of representative, not a feature of the market.

Two guards stop a degenerate history producing a certain quote: move frequencies are shrunk toward 1/3 by one pseudo-count per outcome, a Laplace prior on the trinomial step distribution, so a quiet window cannot freeze the lattice; and variance is floored at a 1bp daily standard deviation. Pricing then proceeds exactly as in Section 3 with the estimates in place of the truth — there is one pricing path, and THEO exercises the same one.

Does it work: what the test can and cannot say

The estimator was scored by drawing parameters, generating a history from them, fitting, and comparing. That tests whether the implementation inverts the process correctly. It does *not* predict live accuracy: the parameter ranges were chosen by hand, and neither the real parameters nor the real warm-up window are knowable in advance. Read the numbers below as “the estimator recovers what it should, and here is the error it carries under those assumptions” — not as a forecast.



Signed error against history length, one panel per quantity, 300 trials with fresh hidden parameters each. Signed rather than absolute, because absolute error cannot distinguish a biased estimator from a merely noisy one. Red marks an interval clear of zero: only κ is ever red, and it is red at every length.

The one real defect

κ is biased upward and is the only quantity that is. Signed error over 300 trials:

history m	30d	120d	1000d
mean signed error	+0.19	+0.046	+0.007
median signed error	+0.09	+0.033	+0.009

The gap between mean and median is a long right tail at short histories. True κ was drawn uniformly from 0.02–0.30, a range chosen by hand, so the *size* of the bias is specific to that choice; its existence and its $1/m$ decay are not.

It is not a coding error — it is a known property of what this estimator is. The rate is an AR(1) process in disguise, meaning tomorrow’s level is a constant fraction of today’s plus noise. To see it, take the expected move from level r : it steps up h with probability $u(r)$ and down h with probability $d(r)$, so

$$\mathbb{E}[\Delta r \mid r] = h(u(r) - d(r)) = h(p_u - p_d) + 2h\kappa(r^* - r),$$

the κ terms adding rather than cancelling because reversion pushes up and down in opposite directions. Rearranging gives $\mathbb{E}[r_{t+1} \mid r_t] = \phi r_t + \text{const}$ with persistence

$$\phi = 1 - 2h\kappa,$$

so estimating κ is estimating an AR(1) slope. That estimator carries Kendall’s classic downward bias in ϕ whenever the intercept is also fitted, because the sample mean of the regressor is built from the same shocks as the errors — and downward in ϕ is upward in κ . Removing the clamp leaves it unchanged, it scales as $1/m$, and it vanishes at very long histories: all consistent with that mechanism and nothing else.

Nor is it harmless. Replacing *only* κ with its true value isolates what it costs: 33–47% of all price error at 30 days, falling to 18–25% at 1000. The mechanism shows in *where* the cost lands. The fit anchors

the rate lines at the mean level of the history, so a slope error costs nothing there and grows with distance — and spot is the *last* value of the window, typically some way from its mean. Splitting on that distance confirms it: at a 15-day expiry and 30 days of history, κ costs 2.7¢ when spot is within one step of the anchor and 5.8¢ beyond it.

Three corrections were tried — a fixed Kendall constant, shrinking the slope, indirect inference — and all three rejected on price error, which is the loss function that matters. Shrinkage is the instructive one: it improves κ 's own accuracy and makes prices *worse*, because it trades noise for a systematic downward tilt, and noise averages out across many contracts while a tilt does not. Scoring a parameter by its own accuracy is the wrong test. The general reason none can work: removing a bias requires estimating it, and that estimate is itself noisy, so at short histories the variance added exceeds the bias removed. This is the uncommon case where a real, confirmed, material bias is left in place deliberately.

What it costs

Median absolute price error by contract family and history length, 60 trials, contracts near the money where they are informative:

median price error	30d	120d	250d	1000d
FED	2.6¢	2.5¢	1.3¢	1.0¢
AJR	6.2¢	3.2¢	2.3¢	1.3¢
THR	5.5¢	4.2¢	3.1¢	1.8¢
spread THR – AJR	6.0¢	3.5¢	3.2¢	1.6¢
weighted $w \cdot \text{AJR} - \text{THR}$	4.3¢	2.6¢	2.2¢	1.2¢

Two readings matter downstream. The rate family is about twice as well priced as the equity families at short histories, so no single half-spread is equally justified across the board. And these are *medians*: the 90th percentile at 120 days is 12–16¢ across every family, so the tail is far heavier than the centre, and any argument leaning on typical error understates a bad draw.

This table is where the report turns. Up to here the question has been how close $\hat{p}$ can get to the truth, and the answer is: within a few cents, with a defect that cannot be economically removed. From here the question is what a price of that quality is *worth* — and the answer turns out to depend far less on the size of the error than on how the quote around it is constructed.

5 The trading problem

Section 4 supplies the parameters. These are fixed within a session, but the values they imply are not: spot levels move and expiry approaches, so $\hat{p}$ is recomputed at every step. A quote is

$$\text{bid} = \hat{p} - W - \text{skew}, \quad \text{offer} = \hat{p} + W - \text{skew},$$

leaving three quantities to be chosen: the half-spread W , the skew, and the size.

Revenue. A round trip buys at $\hat{p} - W$ and sells at $\hat{p} + W$, earning $2W$ regardless of $\hat{p}$, because the estimation error enters both legs and cancels. A market maker is compensated for the spread rather than for accuracy. The cancellation requires both sides to execute.

Information structure. No market price exists. Each maker quotes from its own private estimate and the exchange routes an RFQ to the most aggressive quote, so the winner is the maker whose estimate lies furthest from the field in the direction favourable to the taker. Execution is therefore

concentrated on the side where the estimate is worst — a winner’s curse arising from dispersion among estimators, not from informed order flow. The relevant quantity is thus error *relative to competing makers*, which is unobservable: only one’s own fills are reported, never a competitor’s width and never a lost order.

The spread trade-off. Widening W raises revenue per trade and lowers the fill rate. The objective is capture, $r \cdot W$, where r is the fraction of priced RFQs won.

Outcome variance. A binary payoff has variance $p(1 - p)$ even under perfect knowledge, but a flat book pays payoff – payoff = 0 in every state. Only imbalance carries risk, so outcome variance is a property of inventory, addressed by the skew rather than the spread. Time to expiry enters identically.

Parameter uncertainty. This is a separate quantity from outcome variance and behaves differently. Since $\hat{p}$ is a function of the fitted parameters, $SE(\hat{p})$ measures how far the price would move were the fit re-estimated on other data of the same length. Propagated to price space it is denominated in the same units as $\hat{p}$ and W , so unlike the parameter standard errors themselves it is comparable across contracts. It also differs from $p(1 - p)$ in that it declines as history accumulates, and it is not uniform across the board: it is largest for near-the-money contracts at long expiries and exactly zero for contracts the lattice cannot reach, whose price is already 0 or 1. Its effect on profit nonetheless runs through the same relative channel as every other error, so absolute magnitude is not directly actionable.

Sizing. Margin equals maximum loss, debited on execution and returned only at expiry, so closing a position releases no capital. Selling a contract priced at p locks $1 - p$ per lot: a 5¢ option consumes 95¢ of capacity to sell and 5¢ to buy. Opportunities are therefore ranked by return on capital. A bid of 0.99 on a contract certain to settle at \$1.00 returns about 1%, against 1.9% for an at-the-money quote of the same width.

Channels. An RFQ permits control of size but not of direction. A FOK specifies price and size and admits no partial fill, so there the reservation price acts as an acceptance threshold rather than a size cap. One reservation price must serve both channels. Insolvency scores zero where survival scores partially.

The problem therefore separates. Solvency, sizing and the treatment of certainty are determined by the margin rule; the half-spread is determined by competitors’ quotes, which are not observable.

6 The trading solution

Quantities determined by the rules are computed exactly; quantities determined by competitors are confined to as few parameters as possible.

Solvency. Maximum loss is debited on execution and payoffs credit only at expiry, so remaining within the budget renders insolvency arithmetically impossible. The maker’s ledger must replicate the grader’s, which precludes netting: a ledger that releases capital on close overstates available funds. The ledger is gross, and live quotes reserve capital as executed orders do. One cent is withheld, since $40 - 3 \cdot (40/3)$ evaluates to -3.6×10^{-15} and a negative balance is treated as insolvency.

Sizing. Each order receives the free balance divided by the orders still expected in the step, taken from the previous step’s count. Division by the *remaining* count prevents early arrivals from exhausting the budget: the third of four expected orders receives half the remaining balance rather than one quarter. Both legs of an RFQ are sized at the full allowance while only the larger is reserved, since only one leg can execute.

Refitting. The parameters are constant within a session, so the efficient estimator uses the full

sample: the history is extended by one observation per step and refit on the whole. Measured against known parameters, this removes about 28% of price error at a 30-day burn-in and 12% at 120 days. Rolling and exponentially weighted windows were tested and were uniformly worse, as stationarity predicts. Refitting also corrects the one non-marginal failure mode: the rate step size defaults when the burn-in contains no rate movement, and a single in-session movement identifies it.

The half-spread. W is set to 0.01 with the skew disabled. Two mechanisms were implemented for it, and both are disabled.

A width controller determines W by search. Avellaneda–Stoikov and related models assume a fill intensity $\lambda(\delta) = Ae^{-\kappa\delta}$ decaying smoothly in distance from mid, which admits an interior optimum in closed form. That specification does not hold here. Competitors quote fixed widths, so the probability of winning is a step function of W : within a plateau r is constant, $r \cdot W$ increases linearly, and widening is costless until a competitor is crossed. The controller therefore raises W while capture improves, reverts one step when it does not, and terminates. Steps are multiplicative — W doubles on the way up and halves on the way down, so from the penny floor the ladder is 0.01, 0.02, 0.04, ... — and since doubling W doubles revenue per trade, a step is accepted when it retains more than half its fill rate. Decisions are taken on a fixed number of priced RFQs, and W is clamped to $[0.01, 0.49]$. The search runs separately by underlying, since the field need not quote a single width.

A sizing tilt by $SE(\hat{p})$ was implemented and rejected. At $W = 0.01$ the median contract’s standard error exceeds the edge several times over, so reducing size does not restore a positive expectation. Contracts outside the reachable lattice have $SE = 0$ exactly and consequently attract maximum size, despite carrying the lowest return on capital in the book.